

```
=====
EMPLOYEE SALARY ANALYSIS SYSTEM
=====
```

1. Add Employee
2. Display Employees
3. Selection Sort by Salary
4. Insertion Sort by Salary
5. Search Employee by ID
6. Highest and Lowest Salary
7. Department-wise Display
8. Compare Sorting Algorithms
9. Best/Average/Worst Case Analysis
10. Exit

Enter your choice: 7

Enter Department: IT

Employees in IT department:

```
-----
ID: 101 | Name: Amit | Salary: 30000.00
ID: 103 | Name: Neha | Salary: 40000.00
```

```
=====
EMPLOYEE SALARY ANALYSIS SYSTEM
=====
```

1. Add Employee
2. Display Employees
3. Selection Sort by Salary
4. Insertion Sort by Salary
5. Search Employee by ID
6. Highest and Lowest Salary
7. Department-wise Display
8. Compare Sorting Algorithms
9. Best/Average/Worst Case Analysis
10. Exit

Enter your choice: 6

```
Highest Salary Employee
-----
```

```
ID           : 103
Name          : Neha
Department    : IT
Salary        : 40000.00
```

```
Lowest Salary Employee
-----
```

```
ID           : 102
Name          : Riya
Department    : HR
Salary        : 20000.00
```

```
=====
EMPLOYEE SALARY ANALYSIS SYSTEM
=====
```

1. Add Employee
2. Display Employees
3. Selection Sort by Salary
4. Insertion Sort by Salary
5. Search Employee by ID
6. Highest and Lowest Salary
7. Department-wise Display
8. Compare Sorting Algorithms
9. Best/Average/Worst Case Analysis
10. Exit

Enter your choice: 5

Enter Employee ID to search: 102

Employee Found!

ID	: 102
Name	: Riya
Department	: HR
Salary	: 20000.00

```
=====
EMPLOYEE SALARY ANALYSIS SYSTEM
=====
```

1. Add Employee
2. Display Employees
3. Selection Sort by Salary
4. Insertion Sort by Salary
5. Search Employee by ID
6. Highest and Lowest Salary
7. Department-wise Display
8. Compare Sorting Algorithms
9. Best/Average/Worst Case Analysis
10. Exit

Enter your choice: 8

1. Ascending
2. Descending

Enter required sort order: 1

```
===== SORTING COMPARISON =====
```

```
Selection Sort
Comparisons : 3
Movements   : 4
```

```
Insertion Sort
Comparisons : 2
Movements   : 5
```

```
=====
EMPLOYEE SALARY ANALYSIS SYSTEM
=====
```

1. Add Employee
2. Display Employees
3. Selection Sort by Salary
4. Insertion Sort by Salary
5. Search Employee by ID
6. Highest and Lowest Salary
7. Department-wise Display
8. Compare Sorting Algorithms
9. Best/Average/Worst Case Analysis
10. Exit

Enter your choice: 4

1. Ascending
2. Descending

Enter required sort order: 1

Insertion Sort Result:

ID	Name	Department	Salary
102	Riya	HR	20000.00
101	Amit	IT	30000.00
103	Neha	IT	40000.00

Comparisons : 2

Movements : 5

```
=====
EMPLOYEE SALARY ANALYSIS SYSTEM
=====
```

1. Add Employee
2. Display Employees
3. Selection Sort by Salary
4. Insertion Sort by Salary
5. Search Employee by ID
6. Highest and Lowest Salary
7. Department-wise Display
8. Compare Sorting Algorithms
9. Best/Average/Worst Case Analysis
10. Exit

Enter your choice: 3

1. Ascending
2. Descending

Enter required sort order: 1

Selection Sort Result:

ID	Name	Department	Salary
102	Riya	HR	20000.00
101	Amit	IT	30000.00
103	Neha	IT	40000.00

Comparisons : 3
Movements : 4

```
=====
EMPLOYEE SALARY ANALYSIS SYSTEM
=====
```

1. Add Employee
2. Display Employees
3. Selection Sort by Salary
4. Insertion Sort by Salary
5. Search Employee by ID
6. Highest and Lowest Salary
7. Department-wise Display
8. Compare Sorting Algorithms
9. Best/Average/Worst Case Analysis
10. Exit

Enter your choice: 2

ID	Name	Department	Salary
101	Amit	IT	30000.00
102	Riya	HR	20000.00
103	Neha	IT	40000.00

```
=====
EMPLOYEE SALARY ANALYSIS SYSTEM
=====
```

1. Add Employee
2. Display Employees
3. Selection Sort by Salary
4. Insertion Sort by Salary
5. Search Employee by ID
6. Highest and Lowest Salary
7. Department-wise Display
8. Compare Sorting Algorithms
9. Best/Average/Worst Case Analysis
10. Exit

Enter your choice: 9

```
=====
BEST / AVERAGE / WORST CASE ANALYSIS
=====
```

Record sizes tested: 5, 10, 20, 50

***** BEST CASE *****

Input: Salaries already sorted

N	Sel-Comp	Sel-Move	Ins-Comp	Ins-Move
5	10	0	4	8
10	45	0	9	18
20	190	0	19	38
50	1225	0	49	98

***** AVERAGE CASE *****

Input: Salaries in mixed order

N	Sel-Comp	Sel-Move	Ins-Comp	Ins-Move
5	10	12	7	11
10	45	24	27	36
20	190	48	127	146
50	1225	176	697	746

***** WORST CASE *****

Input: Salaries in reverse order

N	Sel-Comp	Sel-Move	Ins-Comp	Ins-Move
5	10	8	10	18
10	45	20	45	63
20	190	40	190	228
50	1225	100	1225	1323

```
=====
OBSERVATIONS
=====
```

1. Selection Sort performs $O(n^2)$ comparisons in best, average and worst cases.
2. Insertion Sort performs $O(n)$ comparisons in the best case when data is already sorted.
3. Insertion Sort performs $O(n^2)$ operations in average and worst cases.
4. Increasing the number of records increases the number of operations.
5. Selection Sort performs fewer data movements because it performs at most one swap per pass.
6. Insertion Sort is efficient for sorted or nearly sorted data.